1. Foundations, autoencoders and VAEs

Natural logs (nats), except PSNR. State sum vs mean. Greek sigma can mean SD or sigmoid: read the context.

Probability, Gaussian noise and learning

$$p(x, y) = p(x \mid y)p(y), \quad p(x) = \sum_y p(x \mid y)p(y)$$

$$p(y \mid x) = \frac{p(x \mid y)p(y)}{\sum_k p(x \mid k)p(k)}$$

Bayes: likelihood x prior gives a joint score; normalize over classes.

$$\mathbb{E}[f(X)] = \sum_x p(x)f(x), \quad \hat{\mathbb{E}}[f] = \frac{1}{N}\sum_{i=1}^{N} f(x_i)$$

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2, \quad \sigma = \sqrt{v}$$

$$\mathrm{Var}(aX + bE) = a^2\mathrm{Var}(X) + b^2\mathrm{Var}(E)$$

Variance addition here requires independent X and E (or zero covariance).

$$\mathcal{N}(z; \mu, v) = \frac{1}{\sqrt{2\pi v}} e^{-(z-\mu)^2/(2v)}$$

$$\theta \leftarrow \theta - \eta \nabla_\theta L, \quad \frac{\partial L}{\partial w} = \sum_j \frac{\partial L}{\partial u_j}\frac{\partial u_j}{\partial w}$$

Learning rate eta: subtract the gradient; add contributions from all paths.

AE reconstruction, retrieval and latent operations

$$z = f_\phi(x), \quad \hat{x} = g_\theta(z), \quad u = xW + b$$

$$\mathrm{SSE} = \sum_{d=1}^{D} (\hat{x}_d - x_d)^2, \quad \mathrm{MSE} = \frac{\mathrm{SSE}}{D}$$

$$\frac{\partial \mathrm{MSE}}{\partial \hat{x}_d} = \frac{2}{D}(\hat{x}_d - x_d), \quad \text{anomaly if MSE} > \tau$$

$$d(a, b) = \sqrt{\sum_j (a_j - b_j)^2}, \quad \cos(a, b) = \frac{a^T b}{\|a\|\|b\|}$$

Cosine is undefined when either vector is zero. Denoising targets the clean x.

$$z(\lambda) = (1 - \lambda)z_A + \lambda z_B, \quad 0 \le \lambda \le 1$$

$$\Delta z = z_{\text{attribute}} - z_{\text{base}}, \quad z_{\text{edited}} = z_{\text{other}} + \Delta z$$

Image scalar count = H x W x C. Coordinate reduction ratio = input dimension / code dimension; this is not a measured file-compression ratio.

Notation key

x: observation; z: latent code; theta/phi: model/encoder weights; mu: mean; v or sigma squared: variance; sigma: standard deviation; ell: log variance; epsilon: standard Gaussian noise; eta: learning rate; B: batch size; D: data coordinates. The variance identity above is for scalar X.

VAE: distribution, log variance and sampling

$$q_\phi(z \mid x) = \mathcal{N}(\mu, \mathrm{diag}(\sigma^2)), \quad p(z) = \mathcal{N}(0, I)$$

$$\ell = \ln \sigma^2, \quad v = e^\ell, \quad \sigma = e^{\ell/2} = \sqrt{v}$$

$$\varepsilon \sim \mathcal{N}(0, I), \quad z = \mu + \sigma \odot \varepsilon$$

All vector products marked with a circle-dot are elementwise. Generate new data by sampling the prior, then decoding.

KL: direction and the diagonal-Gaussian formula

$$D_{\mathrm{KL}}(P \| Q) = \sum_j P_j \ln \frac{P_j}{Q_j} \ge 0$$

$$D_{\mathrm{KL}}(q \| p) = \mathbb{E}_q[\ln q(z) - \ln p(z)]$$

The first distribution supplies the weights. KL is not symmetric. A zero P term contributes zero; P>0 with Q=0 gives infinity.

$$K = \frac{1}{2}\sum_j \left(\mu_j^2 + v_j - 1 - \ln v_j\right)$$

$$K = -\frac{1}{2}\sum_j \left(1 + \ell_j - \mu_j^2 - e^{\ell_j}\right)$$

Above: q is diagonal Gaussian and p=N(0,I). Check: mu=0 and v=1 gives K=0. Never put a minus sign before the positive-bracket form.

Objective and explicit reductions

$$\mathrm{ELBO} = \mathbb{E}_{q_\phi}[\ln p_\theta(x \mid z)] - K$$

$$L_{\mathrm{VAE}} = \mathbb{E}_{q_\phi}[-\ln p_\theta(x \mid z)] + \beta_{\mathrm{KL}} K$$

$$L_{\mathrm{batch}} = \frac{1}{B}\sum_{i=1}^{B}\left[\frac{1}{D}\sum_d (\hat{x}_{id} - x_{id})^2 + \beta_{\mathrm{KL}} K_i\right]$$

Negative ELBO uses beta_KL=1 and the specified likelihood. The explicit MSE convention above is a practical scaled objective; changing one reduction changes its balance.

$$-\ln p_\theta(x \mid z) = \frac{\|x - \hat{x}\|^2}{2s^2} + \text{constant}$$

For a Gaussian decoder with fixed isotropic observation variance s squared.

Reparameterization and KL gradients

$$\frac{\partial z_j}{\partial \mu_j} = 1, \quad \frac{\partial z_j}{\partial \sigma_j} = \varepsilon_j, \quad \frac{\partial z_j}{\partial \ell_j} = \frac{1}{2}\sigma_j \varepsilon_j$$

$$\frac{\partial K}{\partial \mu_j} = \mu_j, \quad \frac{\partial K}{\partial \ell_j} = \frac{1}{2}(e^{\ell_j} - 1)$$

Hold the sampled epsilon fixed in backpropagation. Add reconstruction and KL gradient paths before updating; apply the stated KL weight and batch reduction. Reconstruction loss depends on the draw; analytic Gaussian KL does not.

Sources: Introduction pp. 3-4, 23-29, 37-41; AE/VAE pp. 5-19, 29-45, 55-64. Page positions refer to the supplied PDFs.

2. GANs, conditional generation and evaluation

GAN expectations use separate real/fake means below. For a combined mean over both sets, the scale can differ.

Adversarial objectives: write the signs first

$$V(D,G) = \mathbb{E}_{x\sim p_{\text{data}}} \ln D(x) + \mathbb{E}_z \ln(1 - D(G(z)))$$

$$\min_G \max_D V(D,G), \quad L_D = -V(D,G)$$

$$L_{G,\text{sat}} = \mathbb{E}_z \ln(1 - D(G(z)))$$

$$L_{G,\text{NS}} = -\mathbb{E}_z \ln D(G(z))$$

Sigmoid, BCE and the Quiz 1 gradient

Per-example derivatives below: a is the discriminator logit and q its sigmoid score. Batch means introduce their averaging factor.

$$q = \sigma(a) = \frac{1}{1+e^{-a}}, \quad \frac{dq}{da} = q(1-q)$$

$$\text{BCE}(y,q) = -y \ln q - (1-y)\ln(1-q)$$

$$\frac{\partial \text{BCE}}{\partial a} = q - y$$

$$\frac{\partial L_{G,\text{sat}}}{\partial a} = -q, \quad \frac{\partial L_{G,\text{NS}}}{\partial a} = q - 1$$

$$\nabla_{\theta_G} L_{G,\text{NS}} = \mathbb{E}_z[(q-1)\nabla_{\theta_G} a]$$

As q approaches 0: logit gradients approach 0 versus -1. D step: detach fakes. G step: freeze D weights but retain the gradient through D.

$$D^*(x) = \frac{p_{\text{data}}(x)}{p_{\text{data}}(x) + p_G(x)}$$

Balanced-prior theory check: if p_G=p_data and D is optimal, D*=0.5. This score alone does not prove matching.

cGAN, StackGAN and dimensions

$$V_c = \mathbb{E}_{x,y} \ln D(x,y) + \mathbb{E}_{z,y} \ln(1 - D(G(z,y),y))$$

$$\hat{c} = \mu(c) + \sigma(c) \odot \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I)$$

Conditioning augmentation uses the same log-variance conversion and Gaussian KL as a VAE. Both cGAN networks receive the condition.

Concatenated input widths add. For fields with K1,...,Km categories: encoding width=sum Kj; combinations=product Kj. StackGAN 64 to 256 multiplies pixel count by 16.

Convolution shape tools: dilation 1

k: kernel size; s: stride; p: padding; C_in/C_out: input/output channels.

$$H_{\text{out}} = \left\lfloor \frac{H_{\text{in}} + 2p - k}{s} \right\rfloor + 1$$

$$H_{\text{out}}^{\text{transpose}} = (H_{\text{in}} - 1)s - 2p + k$$

Transposed formula assumes zero output padding. Apply independently to width.

$$\#\text{conv params} = k_h k_w C_{\text{in}} C_{\text{out}} + C_{\text{out}}$$

Last term is optional bias. Spatial positions do not multiply shared-filter count.

Entropy and Inception Score (higher is preferred)

$$H(P) = -\sum_c P_c \ln P_c, \quad m_c = \frac{1}{N}\sum_{i=1}^{N} P_{ic}$$

$$K_i = \sum_c P_{ic} \ln \frac{P_{ic}}{m_c}, \quad \text{IS} = \exp\left(\frac{1}{N}\sum_i K_i\right)$$

$$\ln \text{IS} = H(m) - \frac{1}{N}\sum_i H(P_i) = I(X;Y)$$

P_ic is the classifier probability for class c on generated image i. Average columns first; sum weighted class terms; average rows; exponentiate once. No extra division by class count.

$$1 \le \text{IS} \le C, \quad P_i = m \ \forall i \implies \text{IS} = 1$$

C is number of classifier classes. Identical confident predictions still give IS=1. IS does not establish within-class novelty.

FID (lower is preferred)

$$\hat{\mu} = \frac{1}{N}\sum_i f_i, \quad \hat{\Sigma} = \frac{1}{N-1}\sum_i (f_i - \hat{\mu})(f_i - \hat{\mu})^T$$

Sample mean and unbiased covariance of features f, for N>1.

$$\text{FID} = \|\mu_r - \mu_g\|^2 + \text{Tr}(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2})$$

$$\text{FID}_{\text{diagonal}} = \sum_j (\mu_{rj} - \mu_{gj})^2 + \sum_j (\sqrt{v_{rj}} - \sqrt{v_{gj}})^2$$

The root is a matrix square root; the diagonal shortcut uses variances v. Trace adds diagonal entries. Matching moments (FID=0) does not prove equality of arbitrary distributions or no copying.

Improved generative precision / recall

$$r_k(u_i; S) = \|u_i - \text{NN}_k(u_i; \{u_j : j \ne i\})\|_2$$

$$F(a; S) = \mathbf{1}\{\exists u \in S : \ \|a - u\|_2 \le r_k(u; S)\}$$

$$\text{Precision} = \frac{1}{|G|}\sum_{g \in G} F(g; R)$$

$$\text{Recall} = \frac{1}{|R|}\sum_{r \in R} F(r; G)$$

R=real features; G=generated features. Build radii within each own set, exclude self by index, and test ANY covering ball. Include boundaries. Quality: precision; coverage: recall.

Image metric and numerical checkpoints

$$\text{PSNR} = 10\log_{10}\frac{\text{MAX}^2}{\text{MSE}}$$

Supporting assignment metric: MAX is the stated peak image value; MSE=0 gives infinite PSNR. State per-image vs pooled averaging.

Quiz IS: 1.406873. Four-image lecture IS: 1.673791. Lecture FID: 2.202041. Lecture 2D precision/recall: 0.8 / 0.6. Keep the examples separate.

Sources: GAN Part 1 pp. 28-55; GAN Part 2 pp. 2-22, 24-96; Quiz 1 pp. 1-3. Shape tools support the worked notes; PSNR is assignment support.

3. Diffusion and DDPM

beta_t = added noise variance; alpha_t = one-step signal share; alpha-bar = cumulative share; epsilon = aggregate forward noise.

Forward process and conditional distribution

$$\alpha_t = 1 - \beta_t, \quad \bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s, \quad \bar{\alpha}_0 = 1$$

$$x_t = \sqrt{\alpha_t}\, x_{t-1} + \sqrt{\beta_t}\, \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, I)$$

$$q(x_t \mid x_{t-1}) = \mathcal{N}(\sqrt{\alpha_t}\, x_{t-1}, \beta_t I)$$

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1 - \bar{\alpha}_t}\, \varepsilon$$

$$q(x_t \mid x_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t}\, x_0, (1 - \bar{\alpha}_t) I)$$

One-shot formula samples a noisy timestep directly. It does not turn reverse generation into one step.

Why the coefficients preserve scale

$$\mathrm{Var}(x_t) = \alpha_t \mathrm{Var}(x_{t-1}) + \beta_t$$

$$\mathrm{Var}(x_t \mid x_0) = (1 - \bar{\alpha}_t) I$$

$$\alpha_2 \beta_1 + \beta_2 = 1 - \alpha_1 \alpha_2$$

Independent Gaussian noises combine by adding variances. Marginal variance stays one only when the previous marginal variance is one; the conditional variance is different.

Recover or match the aggregate noise

$$\varepsilon = \frac{x_t - \sqrt{\bar{\alpha}_t}\, x_0}{\sqrt{1 - \bar{\alpha}_t}}$$

$$\varepsilon = \frac{\sum_{s=1}^{t} \sqrt{\beta_s} \left(\prod_{j=s+1}^{t} \sqrt{\alpha_j} \right) \varepsilon_s}{\sqrt{1 - \bar{\alpha}_t}}$$

This is the aggregate draw corresponding to a particular sequential trajectory. An empty product is 1. A new independent draw matches the distribution, not necessarily the sample.

Noise schedules

$$\beta_t = \beta_{\text{start}} + \frac{t-1}{T-1}(\beta_{\text{end}} - \beta_{\text{start}})$$

$$f(t) = \cos^2\left(\frac{t/T + s}{1 + s} \frac{\pi}{2}\right), \quad s = 0.008$$

$$\bar{\alpha}_t = \frac{f(t)}{f(0)}, \quad \beta_t = \min\left(1 - \frac{f(t)}{f(t-1)}, 0.999\right)$$

After capping beta, recompute the realized cumulative alphas by multiplication. Linear beta does not imply linear signal decay.

Lecture image checkpoint: beta=(0.19,0.36,0.51), alpha-bar_3=0.254016; signal/noise coefficients=(0.504,0.863704). Do not silently clip noisy pixels.

True posterior: condition on clean x0 (t > 1)

$$q(x_{t-1} \mid x_t, x_0) = \mathcal{N}(\tilde{\mu}_t, \tilde{\beta}_t I)$$

$$\tilde{\beta}_t = \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \beta_t$$

$$\tilde{\mu}_t = \frac{\sqrt{\bar{\alpha}_{t-1}}\, \beta_t}{1 - \bar{\alpha}_t} x_0 + \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} x_t$$

Noise prediction gives a reverse sampling rule

$$\hat{x}_0 = \frac{x_t - \sqrt{1 - \bar{\alpha}_t}\, \varepsilon_\theta(x_t, t)}{\sqrt{\bar{\alpha}_t}}$$

$$\mu_\theta(x_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \varepsilon_\theta(x_t, t) \right)$$

$$x_T \sim \mathcal{N}(0, I), \quad x_{t-1} = \mu_\theta(x_t, t) + \sigma_t z$$

$$z \sim \mathcal{N}(0, I) \ (t > 1), \quad z = 0 \ (t = 1)$$

Run t=T down to 1. State the variance choice: commonly sigma_t squared = beta_t or beta-tilde_t. Multiply z by STANDARD DEVIATION sigma_t.

Training: sample x0, random t and epsilon

$$t \sim \mathrm{Uniform}\{1, \ldots, T\}, \quad L_{\text{simple}} = \mathbb{E}_{t, x_0, \varepsilon} \|\varepsilon - \varepsilon_\theta(x_t, t)\|^2$$

$$L_{\text{MSE}} = \frac{1}{D} \sum_j (\hat{\varepsilon}_j - \varepsilon_j)^2, \quad \frac{\partial L_{\text{MSE}}}{\partial \hat{\varepsilon}_j} = \frac{2}{D}(\hat{\varepsilon}_j - \varepsilon_j)$$

Network input=(x_t,t); target=aggregate epsilon. Training does not require a complete reverse trajectory.

Variational bound and why squared noise error appears

$$L_{\text{VLB}} = \mathbb{E}_q \left[L_T + \sum_{t=2}^{T} L_{t-1} + L_0 \right]$$

$$L_T = D_{\text{KL}}(q(x_T \mid x_0) \| p(x_T)), \quad L_0 = -\ln p_\theta(x_0 \mid x_1)$$

$$L_{t-1} = D_{\text{KL}}(q(x_{t-1} \mid x_t, x_0) \| p_\theta(x_{t-1} \mid x_t))$$

$$D_{\text{KL}}(\mathcal{N}(m_q, s^2 I) \| \mathcal{N}(m_p, s^2 I)) = \frac{\|m_q - m_p\|^2}{2s^2}$$

$$L_{t-1} = C_t + w_t \|\varepsilon - \varepsilon_\theta\|^2, \quad w_t = \frac{\beta_t^2}{2\sigma_t^2 \alpha_t (1 - \bar{\alpha}_t)}$$

Last line: theta-dependent mean term for fixed reverse variance; C_t is theta-independent. Dropping w_t changes weighting across timesteps, yielding the simplified objective.

Sources: Diffusion Part 1 pp. 14-34; Part 2 pp. 4-39 and Appendix A pp. 43-46. Fixed-variance DDPM conventions are stated below.

4. Transformers, attention and sequence losses

Use row vectors consistently. Softmax runs across keys; positional angles use radians. N = non-padding target count.

Embeddings and attention: the five-step calculation

d_model: embedding width; d_k: key/query width; d_v: value width; h: heads. m/n: query/key token counts.

$$X_0 = \sqrt{d_{\text{model}}}\, E[\text{token IDs}] + \text{PE}$$

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V$$

$$S = \frac{QK^T}{\sqrt{d_k}} + M, \quad A = \text{softmax}_{\text{rows}}(S), \quad O = AV$$

$$A_{ij} = \frac{e^{S_{ij}-m_i}}{\sum_k e^{S_{ik}-m_i}}, \quad m_i = \max_j S_{ij}$$

$$o_i = \sum_j A_{ij} v_j, \quad \sum_j A_{ij} = 1$$

Project -> dot products -> scale -> mask/softmax -> weighted VALUES. Output is a contextual vector, not yet a vocabulary probability.

Masks, self-attention and cross-attention

$$M_{ij} = 0 \ (j \le i), \quad M_{ij} = -\infty \ (j > i)$$

Apply the causal mask BEFORE softmax. Self-attention: Q/K/V from the same sequence. Cross-attention: Q from decoder, K/V from encoder. Source padding masks remain separate.

$$Q : m \times d_k, \ K : n \times d_k, \ V : n \times d_v$$

$$S : m \times n, \quad O : m \times d_v$$

Self-attention has m=n. Per-head score count=n squared; h heads give h n squared scores.

Multi-head, feed-forward, residual and LayerNorm

$$\text{head}_r = \text{Attention}(XW_Q^r, XW_K^r, XW_V^r)$$

$$\text{MHA}(X) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h) W_O$$

$$\text{FFN}(x) = \text{ReLU}(xW_1 + b_1)W_2 + b_2$$

$$y = \text{LayerNorm}(x + \text{Sublayer}(x))$$

$$\mu_u = \frac{1}{d}\sum_j \mu_j, \quad v_u = \frac{1}{d}\sum_j (u_j - \mu_u)^2$$

$$\text{LN}(u)_j = \gamma_j \frac{u_j - \mu_u}{\sqrt{v_u + \varepsilon_{\text{num}}}} + b_j$$

Lecture uses post-norm. Normalize across one token's feature coordinates. Gamma and b are learned gain and shift; epsilon_num is a small numerical stabilizer.

Standard equal-width heads: d_k=d_v=d_model/h. Concatenate h outputs of width d_v; do not average them. Residual shapes must match.

Sinusoidal positional encoding

$$\text{PE}(p, 2r) = \sin\left(\frac{p}{10000^{2r/d_{\text{model}}}}\right)$$

$$\text{PE}(p, 2r+1) = \cos\left(\frac{p}{10000^{2r/d_{\text{model}}}}\right)$$

Zero-based coordinate pairs r. A pair shares one denominator; interleave sine/cosine. At position 0, pairs are (0,1).

$$\sin a \sin b + \cos a \cos b = \cos(a - b)$$

$$(c_i + p_i)^T (d_j + r_j) = c_i^T d_j + c_i^T r_j + p_i^T d_j + p_i^T r_j$$

The pure positional identity gives an offset signal. Full projected attention also contains content terms; longer-length performance is not guaranteed by computable codes.

Vocabulary probabilities and cross-entropy

$$a_t = h_t W_{\text{vocab}} + b, \quad p_{tv} = \frac{e^{a_{tv}}}{\sum_u e^{a_{tu}}}$$

$$\text{CE}_t = -\sum_v y_{tv} \ln p_{tv} = -\ln p_{t,\text{true}}$$

$$L_{\text{sum}} = \sum_{t=1}^{N} \text{CE}_t, \quad L_{\text{mean}} = \frac{L_{\text{sum}}}{N}$$

$$\frac{\partial \text{CE}_t}{\partial a_{tv}} = p_{tv} - y_{tv}$$

One-hot true targets, not argmax probabilities. Mean token loss divides each gradient by N. Exclude padding and include EOS when supplied.

Teacher forcing: inputs [SOS,y1,...,y_(N-1)] predict targets [y1,...,yN]. During inference use the generated prefix.

Autoregressive likelihood, greedy and beam

$$P(y_{1:N} \mid x) = \prod_{t=1}^{N} p(y_t \mid y_{<t}, x)$$

$$\text{score}(y_{1:N}) = \sum_t \ln p(y_t \mid y_{<t}, x)$$

$$y_t^{\text{greedy}} = \arg\max_v p(v \mid y_{<t}, x)$$

Beam keeps the best B expanded prefixes by cumulative score. It is approximate; state length conventions when comparing different lengths.

Parameter counts and checked lecture answers

$$\#\text{embedding params} = |\mathcal{V}| d_{\text{model}}$$

$$\#\text{vocab-head params} = d_{\text{model}}|\mathcal{V}| + |\mathcal{V}|$$

Vocabulary-head count includes bias; tied embedding/head weights change the distinct parameter count.

Robot output: (5.606850, 2.187793), correcting the original slide arithmetic. Cross-attention output: (3.748287, 1.465033). Sentence CE sum=6.543112; mean=1.308622.

Sources: Transformer pp. 11-29, 30-58, 59-67; corrected lecture numericals in the approved handwritten notes. No actual midterm blueprint was supplied.